AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-004	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF THE ASH CONTENT (COAL-QUICK ASH METHOD)		

Assessment Ash Content

Date: 06/11

Candidate name: Thapo **Staff No:** 04

Candidate Signature: Thapo

No	Question:	Points	Score
1.	At what temperature is the ash sample incinerated to constant mass? 815°C ✓	* 1	1
2.	What problems could be encountered if the sample is incinerated in a furnace without a specified ramping profile? Cripples validity of results as unspecified ramping = Uncalibrated uncertified setpoints	1	1
3.	Between which masses are the ash samples controlled? 0,2490 – 1,05 (0.5000)	2	1
4.	What is the maximum temperature at which samples can be inserted into the furnace? I.e ISO ash = 100% A.Ash = 825°C	2	2
5.	Complete the sentence: The sample is incinerated in a muffle furnace at 818°C to determine the amount of ash within a coal sample in percentage	* 2	2
6.	Where are the crucibles cooled after the analysis is complete? Metal plate & desiccator (optional)	* 2	2
7.	What accompanies each batch of samples? CRM ✓	1	1
8.	What is the maximum number of crucibles allowed in a furnace? ≈ 20 ✓	* 1	1

9.	Who is responsible for investigating any deviation? Initially lab tech with lab supervisor/manager guidance.	* 1	
10.	Two grams of samples can be used for ash. True or False? True	1	
11.	The repeatability of ash is 0.3%. True or False? False	1	
12.	The unit for reporting ash is grams True or False? False	* 1	
13.	What is the formula for calculating the ash content? Ash = Final mass − Initial mass / sample mass	1	
14.	Who is responsible for plotting control charts? Lab technician	* 1	
17.	A single repeat sample can be used for ash True/False False	* 1	
18.	At what temperature are the ash crucibles weighed back? Room temperature	1	
19.	What is the repeatability limit for ash determinations? 0,2 for samples <10% 2% ave >10	* 1	
20.	Unmarked and unidentified crucibles can be used for ash determination. True/ False False	* 1	
	Total:	23	22

Comments by Assessor(s): Competent

Comments by Candidate:

Assessor Declaration:

I hereby state that the candidate has been assessed according to the requirements of this document and found to be *competent / not yet competent** **and** *may / may not* proceed* **to the Assessment Panel for authorisation.**

* Delete as applicable.

Assessor Name N. Mkhal **Signature:** _____ **Date:** 6/11/2025

If you do not agree with the feedback, please discuss this. If you cannot come to an agreement, ask for an explanation of the appeals procedure.

I declare that I am satisfied with this assessment, and I can perform these tasks.

Learner Signature: _____ Date: 06/11

Senior analyst/supervisor Signature: _____ Date: _____

Learner Signature: _____ Date: 06/11

Note: 1. This shall be done after theoretical test has been completed.
2. Trainee to be constantly supervised until Initial training is signed off by competent trainer.

Sample Used		Lims Number	Known Value	Duplicate Test Results	
Sept 21		1610	21.60	21.06	21.07

Acceptance Tolerance: 0.2% between duplicate results for Ash≤10% and 2% of average for ash>10%.

AzurESky "ensure quality of test"	AZUREKY PTY LTD	Document # AZCA-003	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF CALORIFIC VALUE IN COAL		

PERSONAL DETAILS:

Name:	Maponya TMJ
Staff No:	04
Job Title:	Laboratory Supervisor

ASSESSMENT:

Attempt:	01
Date:	07/11/2025
Time:	10:00

ASSESSORS:

Name & Staff nr.:	1. SB Nyamubi	
	2. N. Msibi	
	3.	
	4.	
Observers: (optional)	1.	2.
Translator: (if required)		

AzurESky "ensure quality of test"	AZUREKY PTY LTD	Document # AZCA-003	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF CALORIFIC VALUE IN COAL		

Instructions and Information

1.	The competency-based assessment instrument was developed in line with the requirements to be declared competent in **Determination of the Calorific Value in Coal** **Desired Outcome:** At the end of this module the employee should be able to perform total sulphur determination in line with the test procedure in a manner which will give accurate, repeatable and traceable results. The employee should have adequate knowledge of the method and its limitations
2.	The competency and development tool is based on ISO 1928
3.	You are allowed to use the following documentation during the assessment: • Any manual(s) associated with the Calorific Value determination(s) in use. • Any Corporate Standard, Corporate Guideline or Corporate Manual. • Any other relevant **printed** reference material you may feel will assist you.
4.	You are going to demonstrate the determination of Calorific Value. During the assessment, the assessor may ask you additional relevant questions, the answer to which he/she will then immediately note down on the assessment sheet. The assessment consists of **23** points of which **14** are critical. To be declared competent, you must obtain 80% for the assessment. You are not allowed to get one of the critical questions wrong. The assessment shall then be stopped, and the result will be that you are not yet competent. Some of the questions may be structured to assess your reflective competence. There is a set time limit of **60** minutes for the completion of the assessment
5.	You have the right to an interpreter and the right to have an observer present during the assessment.
6.	You are allowed to ask the assessor(s) to clarify any question during the assessment.

AzurESky "ensure quality of test"	AZUREKY PTY LTD	Document # AZCA-003	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF CALORIFIC VALUE IN COAL		

Assessment of Calorific Value

Date: 07/11

Candidate name: Thepo Staff No: 04

Candidate Signature: [signature]

No	Question:	Points	Score
1.	What is meant by calorific Value? Measurement of the heating capacity of coal	* 1	
2.	What chemical is used to calibrate the machine? Benzoic acid pill	* 1	
3.	What is meant by repeatability? The accepted difference between original and duplicate analyzed same operator, same lab, same time	1	
4.	What is used to clean the bomb vessel inside? soap, water and a brush	1	
5.	Can grease or oil be used on oxygen gauges? No	1	
6.	Complete the sentence: Duplicate results by the same analyst, same equipment, short span of time	* 2	
7.	Which documents must be used to record raw data? CV raw data recording form	* 1	
8.	How often is the instrument calibrated? *When there is a defect *instrument drift i.e control chart bias; control out of limits *If a major/critical element of the system went for repairs *If the equipment is new	1	

9.	How often are and with what are drift checks analyzed? ① Drift checks are done initially ② After 10 runs by CRM, then Benzoic acid when a drift is detected.	* 2	
10.	Can the bombs be immersed in water? NO	1	
11.	A Benzoic Acid can be used until the bottle is empty True or False? True	* 1	
12.	Stanley knife is used to cut o-rings. True or False? True	1	
13.	Within which masses are test samples controlled 0,4975 – 0,5025	* 2	
14.	What is the repeatability limit for Calorific Value determination? ~~10 Tms~~ < 0,12	* 1	
15.	When is the maintenance on Calorific Value done? Before each use	* 1	
16.	How many samples are used in repeat analyses? 1 (duplicate)	* 1	
17.	Who plots the control charts? Operator who analyzed for the day	* 1	
18.	What material is used to clean electrodes? Sandpaper	1	
19.	Calorific value crucibles can never be reused True/ False False	1	
20.	What gas is used in the calorific value analyzer? Oxygen	1	

	Total:	23	23

Comments by Assessor(s): Competent

Comments by Candidate:

Assessor Declaration:

I hereby state that the candidate has been assessed according to the requirements of this document and found to be *competent / not yet competent and *may / may not* proceed* to the Assessment Panel for authorisation.**

Assessor Name N. Mold **Signature:** _____ **Date:** 7/11/2025

Learner Signature: _____ **Date:** 07/11

Note: 1. This shall be done after theoretical test has been completed.
2. Trainee to be constantly supervised until Initial training is signed off by competent trainer.

Sample Used		Lims Number	Known Value	Duplicate Test Results	
Sept P1		1610	25.35	25.27	25.34
Acceptance Tolerance: 0.12% between duplicate results					

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-002	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF VOLATILE MATTER		

PERSONAL DETAILS:

Name:	Mapanya TMJ
Staff No.:	04
Job Title:	Laboratory Supervisor

ASSESSMENT:

Attempt:	01
Date:	07/11/2025
Time:	10:00

ASSESSORS:

Name & Staff nr.:	1. M. Msibi	
	2.	
	3.	
	4.	
Observers: (optional)	1.	2.
Translator: (if required)		

Instructions and Information

1.	The competency-based assessment instrument was developed in line with the requirements to be declared competent in **Determination of Volatile Matter in Coal** **Desired Outcome:** At the end of this module the employee should be able to perform volatile matter determination in line with the test procedure in a manner which will give accurate, repeatable and traceable results. The employee should have adequate knowledge of the method and its limitations
2.	The competency and development tool is based on ISO 562
3.	You are allowed to use the following documentation during the assessment: • Any manual(s) associated with the Volatile Matter determination(s) in use. • Any Corporate Standard, Corporate Guideline or Corporate Manual. • Any other relevant **printed** reference material you may feel will assist you.
4.	You are going to demonstrate the use of determination of the Volatile Matter. During the assessment; the assessor may ask you additional relevant questions, the answer to which he/she will then immediately note down on the assessment sheet. The assessment consists of **27** points of which **24** are critical. To be declared competent, you must obtain 80% for the assessment. You are not allowed to get one of the critical questions wrong. The assessment shall then be stopped, and the result will be that you are not yet competent. Some of the questions may be structured to assess your reflective competence. There is a set time limit of **60** minutes for the completion of the assessment
5.	You have the right to an interpreter and the right to have an observer present during the assessment.
6.	You are allowed to ask the assessor(s) to clarify any question during the assessment.

Assessment for Volatile Matter

Date: 01 / 11

Candidate name: Thup **Staff No:** 04

Candidate Signature: [signature]

No	Question:	Points	Score
1.	At what temperature is the test sample incinerated? 900°C ± 8°C	* 1	
2.	What does one use to monitor the time in the furnace? Stopwatch / Timer	* 1	
3.	What is the maximum time for incineration of the test sample? 7 minutes ± 2seconds.	* 1	
4.	To what temperature must the volatile furnace recover after 4 minutes? 900 ± 8°C (896°C)	* 1	
5.	Where does one cool the crucible and stands? Metal Plate	1	
6.	Complete the sentence: If the furnace has not recovered the sample is disregarded and a new set is prepared for a repeat.	* 2	
7.	What is meant by repeatability? Acceptable limits between an origional and a duplicate of a sample as set out in a certain parameter.	* 3	
8.	How often drifts are checks run? Drifts are checked by analysis of a CRM that accompanies a sample each time when there is an analysis requested.	* 1	

9.	How often and by whom are the verification checks done? Vegication are done weekly	* 2	
	by competent lab technicians.		
10.	Between which masses are the masses for volatile determination controlled?	* 1	
	0,95 - 1.05		
11.	The insertion time of a volatile stand can be 10 seconds. True or False?	* 1	
	False.		
12.	A volatile crucible stand can be kept in the furnace for 7min± 8 seconds. True or False?	* 1	
	False, though MV prved stable even at 8min.		
13.	How does one identify volatile crucible and lids? Lid & crucibles are to have the	* 2	
	same number		
14.	What is the repeatability limit for volatile determination? 0,3 <10%; 3% abe >10%.	* 1	
15.	What is subtracted after testing to get volatile results?	* 1	
	Inherent moisture.		
16.	How many samples must be used in the repeat analyses?	1	
	A (duplicate)		
17.	How often are crucibles cleaned? Every day before initial usage.	* 1	
18.	What should the distance between the crucible and the lid?	1	
	0,5mm		
19.	How does one ensure that new lids fit their crucible? The distance between the lid	* 1	
	& crucible must be less than 0,5mm		
20.	What laboratory documents are to be used when	*	

	determining the Volatile Matter Temperature recovery form	1	
21	Are you allowed to carry out volatile analysis with vacant places in your stand?	* 2	
	Never.		
	Total:	27	27

Comments by Assessor(s): Competent

Comments by Candidate:

Assessor Declaration:

I hereby state that the candidate has been assessed according to the requirements of this document and found to be *competent / not yet competent and *may / may not* proceed* to the Assessment Panel for authorisation.**

Assessor Name N. Mabi **Signature:** _____ **Date:** 7/11/2025

Learner Signature: _____ **Date:** 07/11

Note: 1. This shall be done after theoretical test has been completed.
2. Trainee to be constantly supervised until Initial training is signed off by competent trainer.

Sample Used		Lims Number	Known Value	Duplicate Test Results	
Sept PT		1610	23.66	24,13	24,11
Acceptance Tolerance: 0.3% or 4% of average between duplicate results					

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-001	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

PERSONAL DETAILS:

Name:	Maponya TMJ
Job Title:	Laboratory Supervisor

ASSESSMENT:

Attempt:	01
Date:	07/11/2025
Time:	10:00

ASSESSORS:

Name	1. N. Msibi	
	2.	

AzurESky	AZURESKY PTY LTD	Document # AZCA-001	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

Instructions and Information

1.	The competency-based assessment instrument was developed in line with the requirements to be declared competent in **the Determination of Inherent Moisture** **Desired Outcome:** At the end of this module the employee should be able to perform Determination of inherent Moisture in line with the test procedure in a manner which will give accurate, repeatable and traceable results. The employee should have adequate knowledge of the method and its limitations
2.	You are allowed to use the following documentation during the assessment: • Any manual(s) associated with Inherent Moisture determination(s) in use. • Any Corporate Standard, Corporate Guideline or Corporate Manual. • Any other relevant **printed** reference material you may feel will assist you.
3.	You are going to demonstrate the determination of the inherent moisture During the assessment, the assessor may ask you additional relevant questions, the answer to which he/she will then immediately note down on the assessment sheet. The assessment consists of **26** points of which **16** are critical (*). To be declared competent, you must obtain 80% for the assessment. You are not allowed to get one of the critical questions wrong. The assessment shall then be stopped, and the result will be that you are not yet competent. Some of the questions may be structured to assess your reflective competence.
4.	You have the right to an interpreter and the right to have an observer present during the assessment.
5.	You are allowed to ask the assessor(s) to clarify any question during the assessment.

AzurESky	AZURESKY PTY LTD	Document # AZCA-001	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

Assessment Inherent Moisture

Date: 01/11

Candidate name: Tshepo **Staff No:** 04

Candidate Signature: Mpanya

No	Question:	Points	Score
1.	At what temperature is the test sample analysed? 107 °C ✓	* 1	1
2.	Who is responsible for plotting control charts? Operator/analyst testing for moisture for the day/shift.	* 1	1
3.	Who is responsible for investigating any deviation? Initially lab tech, guided by supervisor/ manager	* 1	1
4.	What is meant by the term repeatability? Acceptable limit between an original and a duplicate done by same operator, instrument under 1 sitting.	2	1
5.	What does high inherent moisture content indicate? The coal water content is high.	1	1
6.	Complete the sentence: Oxidized Coals become porous when surface area is increased which allows for movement of oxygen which increases the likelihood of combustion.	* 2	2
7.	Where must the crucibles be cooled and for how long? 10 minutes outside the oven/metal plate, 10 minutes inside dessicator	* 2	
8.	What accompanies each batch of samples for inherent moisture? CAM ✓	* 1	1
9.	How must the crucible lid be? When inside oven, placed on top but open, close at the oven door when removing from oven	1	

AzurESky "ensure quality of test"	AZURESKY PTY LTD	Document # AZCA-001	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

10.	Name the one desiccant that can be used?	1	
	Silica gel		
11.	The inherent moisture sample is incinerated for one hour. True or False?	* 1	
	False		
12.	The inherent moisture crucible is weighed back while still hot. True or False?	* 1	
	False		
13.	Inherent moisture crucibles are cooled with the lid open True or False?	* 1	
	False		
14.	What is the function of silica gel in the dessicators?	1	
	Absorb excess moisture.		
15.	How does one regenerate silica gel?	1	
	Introduce into the oven.		
16.	How many samples must be used in the repeat analyses?	1	
	1 (duplicate)		
17.	Within which limits must mass be controlled? 0,95 – 1,05	* 1	
18.	What are the repeatability limits for inherent moisture? TMS	* 4	
	0,1 < 0,5% 0,15 > 5%		
19.	Inherent moisture can only be done when the balance has been verified True/ False	* 1	
	True		
20.	What is the relation between inherent moisture and volatile matter? TMS	1	
	~~The higher~~ Inversly proportional		
	Total:	26	26

AzurESky	AZURESKY PTY LTD	Document # AZCA-001	Revision# 0
Subject	ASSESSMENT FOR DETERMINATION OF MOISTURE CONTENT OF COAL IN THE ANALYSIS COAL SAMPLE		

Comments by Assessor(s): Competent

Comments by Candidate:

Assessor Declaration:

I hereby state that the candidate has been assessed according to the requirements of this document and found to be *competent / not yet competent and *may / may not* proceed* to the Assessment Panel for authorization.**

Assessor Name N. Mabi **Signature:** _____ **Date:** 7/11/2025

Learner Signature: _____ Date: 07/11/25

Note: 1. This shall be done after theoretical test has been completed.
2. Trainee to be constantly supervised until Initial training is signed off by competent trainer.

Sample Used		Lims Number	Known Value	Duplicate Test Results	
Sept F1		1610	1.82	1.77	1.77
Acceptance Tolerance: 0.15% between duplicate results for IM >5% and 0.1 for IM ≤5%					